Prospective study of the association between grapefruit intake and risk of breast cancer in the European Prospective Investigation into Cancer and ...

Grapefruit inhibits cytochrome P450 3A4 and may affect estrogen metabolism. In the European Prospective Investigation into Cancer and Nutrition (EPIC), we examined the relationships of grapefruit intake with risk of breast cancer and with serum sex hormone levels. 114,504 women with information on dietary intake of grapefruit and on reproductive and lifestyle risk factors were followed for a median 9.5 years and 3,747 incident breast cancers were identified. Fifty-nine percent of women reported eating grapefruit, 4% ate > or = 60 g/day. Cox proportional hazard models were used to estimate the hazard ratio (HR) for breast cancer according to grapefruit intake, adjusting for study centre, reproductive factors, body mass index, energy intake, and alcohol intake. Grapefruit intake was not related to the risk of breast cancer: compared with women who ate no grapefruit, women with the highest intake of > or =60 g/day had a HR of 0.93 (95% CI 0.77-1.13), p for linear trend = 0.5. There was no relationship between grapefruit intake and breast cancer risk among premenopausal women, all postmenopausal women, or postmenopausal women categorized by hormone replacement therapy use (all p>0.05). There was no association between grapefruit intake and estradiol or estrone among postmenopausal women. In this study, we found no evidence of an association between grapefruit intake and risk of breast cancer.